

Course Title: Primary Metabolic
Pathways III- Metabolism of
Amino Acids and Proteins

Course Code : BCH 353



Topic:
Amino Acid Composition of Proteins

Course Lecturer:

Innocent U. Okagu

[innocent.okagu@unn.edu.ng]

Biochemistry old block- Prefab RM 115



Learning Objectives

When you have mastered this section of the course, you should be able to complete the following objectives:

- Define amino acids.
- List the key properties of amino acids.
- Draw the structure of a typical amino acid and indicate the following features, which are common to all amino acids: basic functional groups, side chains and α - carbon.

Learning Objectives Contd.

- Classify each of the 20 common amino acids according to the side chain on the α -carbon, nutritional needs and draw the structure of at least, one amino acid in each class.
- Give the name, one-letter and three-letter symbols of each amino acid.
- Explain how amino acid side chains and their interactions influence the nature and function of proteins.

Recommended Textbooks

- **Lehninger Principles of Biochemistry**, 5th Edition by David L. Nelson and Michael M. Cox.
- **Biochemistry**, 5th Edition by Jeremy M. Berg, John L. Tymoczko and Lubert Stryer.
- **Biochemistry** by Garrett R.H and Grisham C.M.
- **BIOS Instant Notes in Biochemistry**, 4th Edition by David Hames and Nigel Hooper.

Amino Acid Components of Proteins

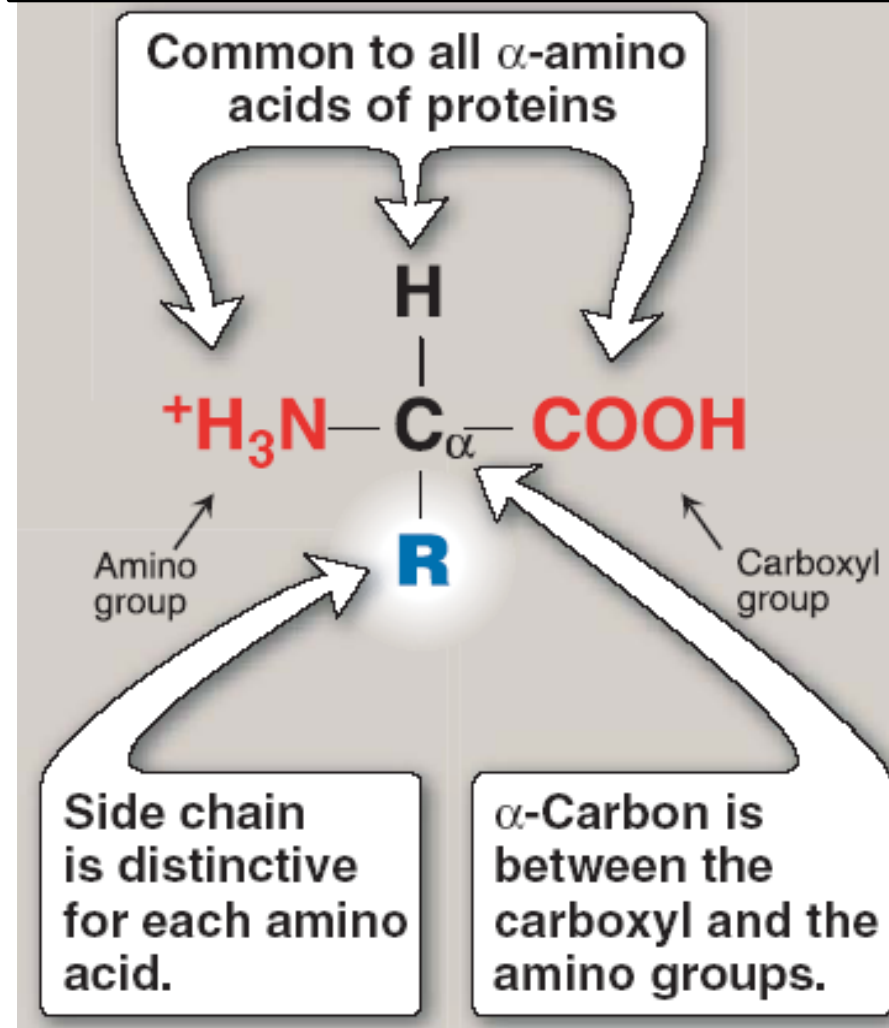
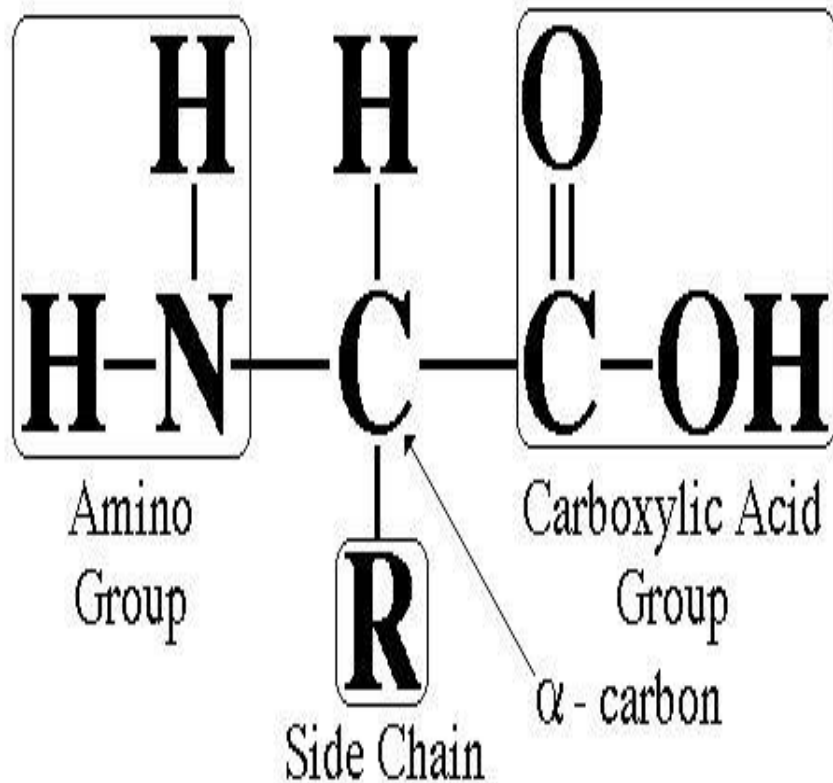
- Amino acids are the monomeric units, or building blocks of proteins joined by a specific type of covalent linkage.
- Properties of proteins depend on the characteristic sequence of component amino acids, each of which has distinctive side chains.

Amino Acids as Sources of Nitrogen

- Almost all nitrogen-containing compounds of the body are synthesized from amino acids—
 - ✓ Cellular proteins (eg. Collagen, albumin)
 - ✓ Hormones (e.g. thyroxine, glucagon, and insulin),
 - ✓ Enzymes
 - ✓ Catecholamines and neurotransmitters,
 - ✓ Haem (and globin) in haemoglobin and cytochromes,
 - ✓ Melanin (dark pigments of the skin),
 - ✓ Purine and pyrimidine bases

AA Components of Proteins Contd.

- Amino acids are produced by digestion of dietary proteins by proteases:
 - ✓ Stomach (pepsin),
 - ✓ Pancreas (trypsin, chymotrypsin, elastase and carboxypeptidases), and
 - ✓ Intestine (enteropeptidase and aminopeptidases),
- The amino acids produced are absorbed through the intestinal epithelial cells, and enter the blood.



Structure of a typical Amino acid

Properties of Amino Acids

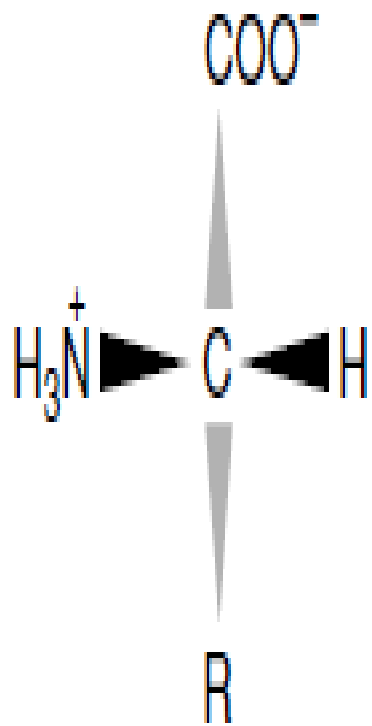
Amino acids generally possess the following properties:

- (1) Optical properties or chirality (except glycine)
- (2) Acid–base properties,
- (3) The capacity to polymerize, and
- (4) Varied structure and chemical functionality in the amino acid side chains

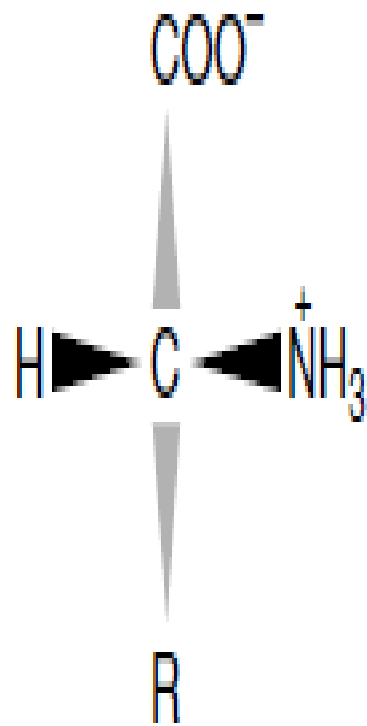
The properties of the amino acid vary as the structure of R varies.

Chirality of Amino Acids

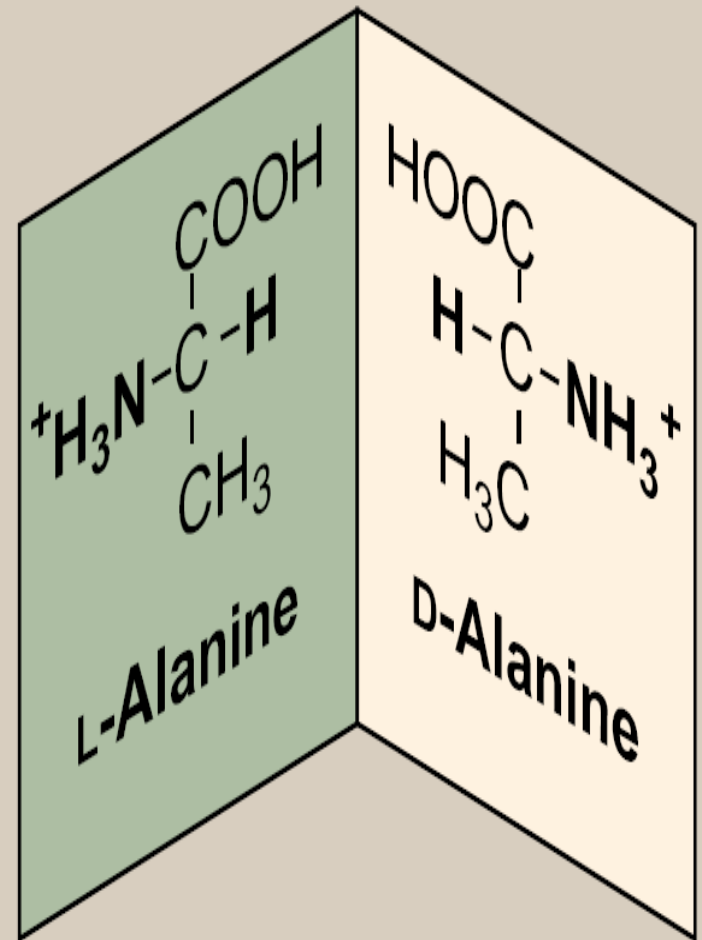
- The α -carbon of AA attached to four different chemical groups is a chiral or optically active carbon atom.
- All AAs are chiral (except glycine).
- Amino acids exist in two forms, D and L, that are mirror images of each other.
- All amino acids found in proteins are of the L-configuration.



L-Amino Acid

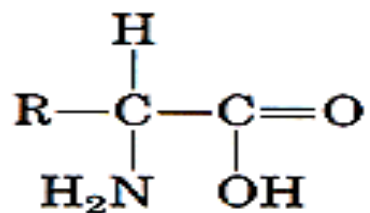


D-Amino acid

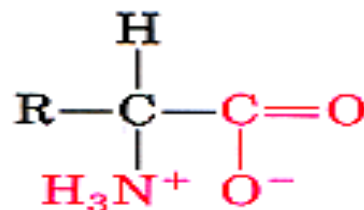


Enantiomeric nature of some amino acids

Acid-Base Properties of amino acids

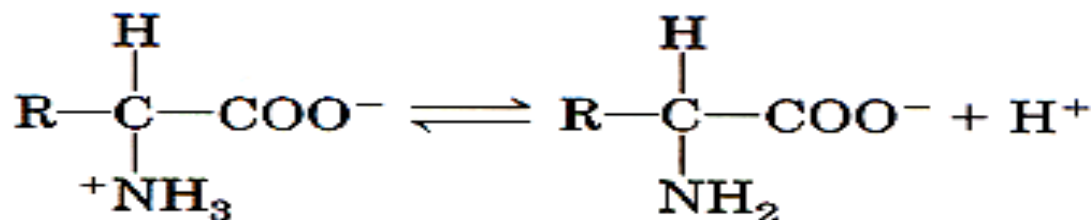


Nonionic form



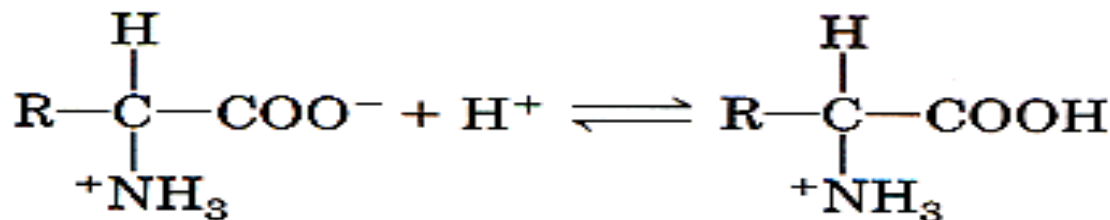
Zwitterionic form

Zwitterion form



Zwitterion
as acid

Weak acid

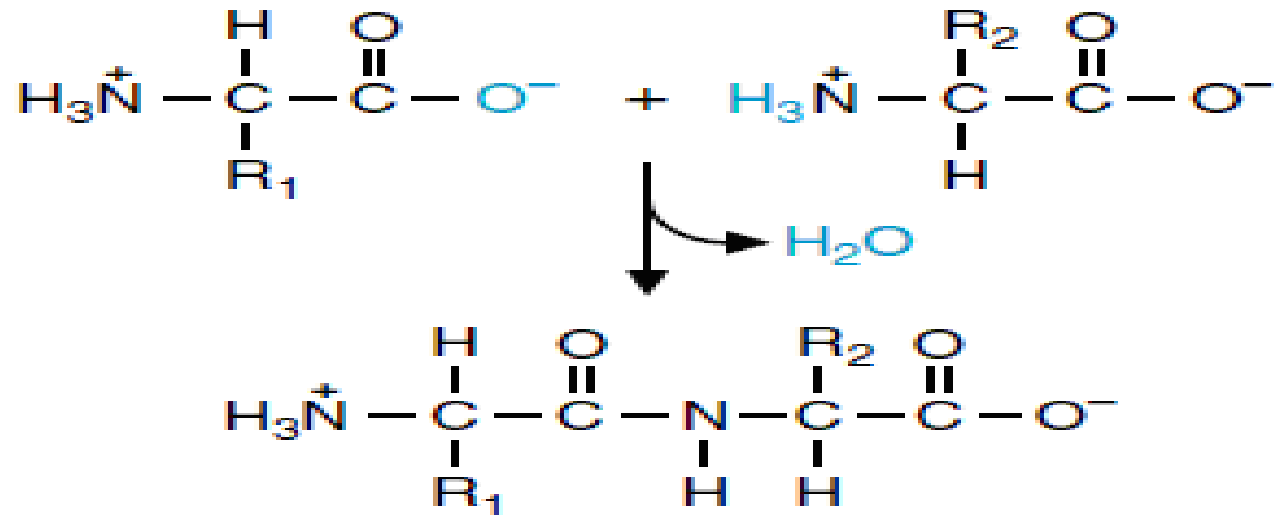


Zwitterion
as base

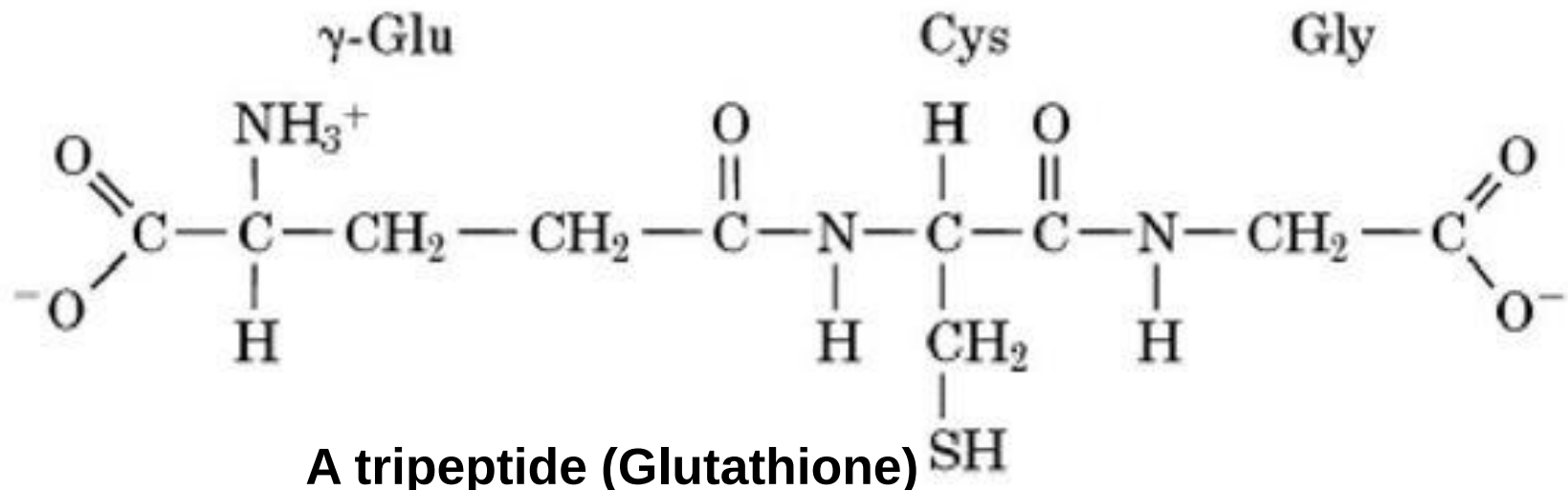
Weak base

The zwitterion predominates at neutral pH

Capacity to Polymerize

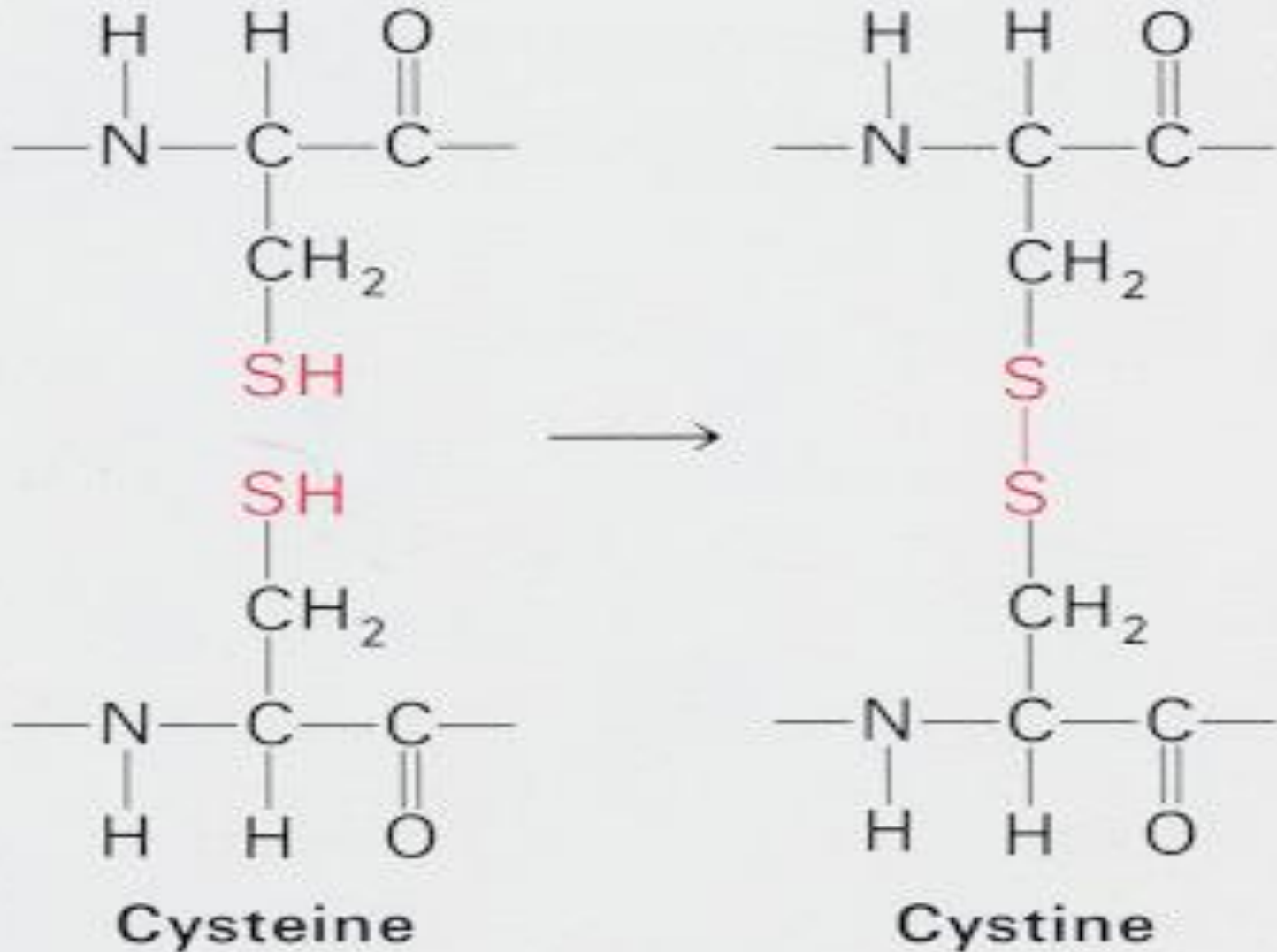


A dipeptide

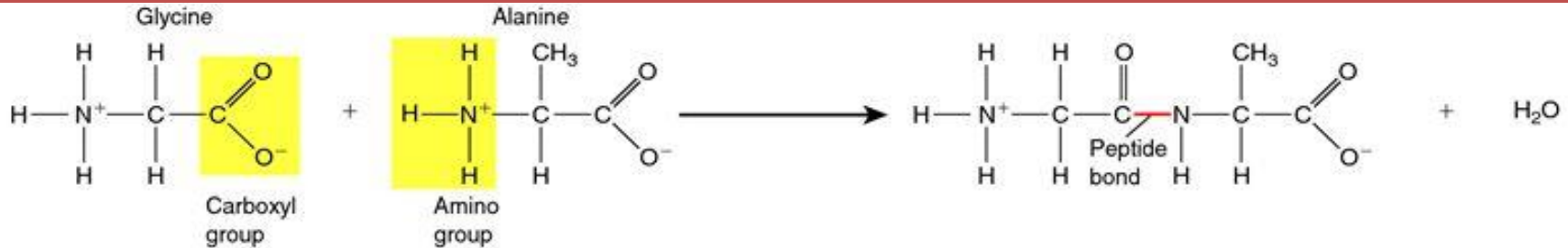


A tripeptide (Glutathione)

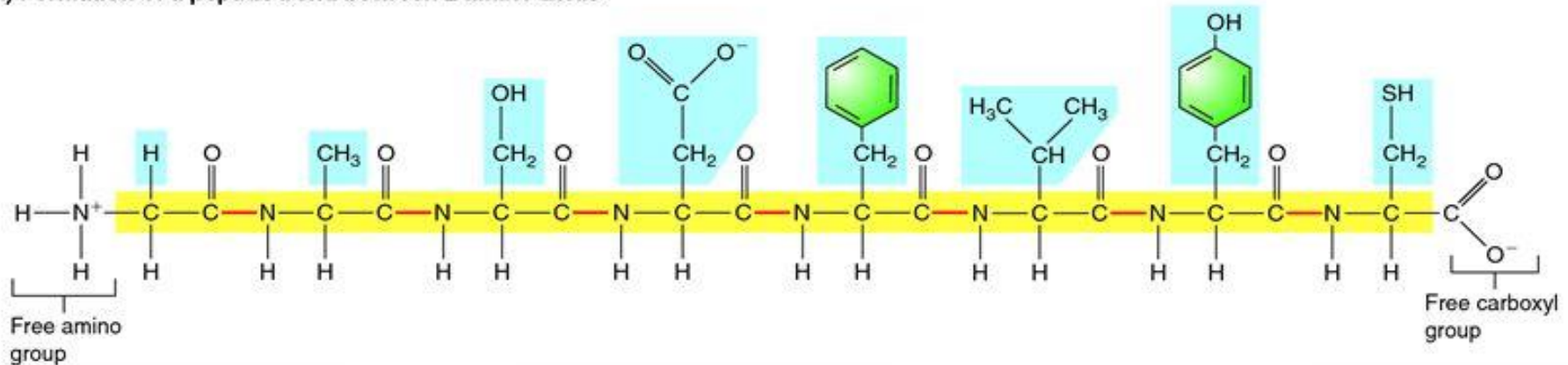
Formation of Disulphide Bond



Formation of Polypeptide



(a) Formation of a peptide bond between 2 amino acids

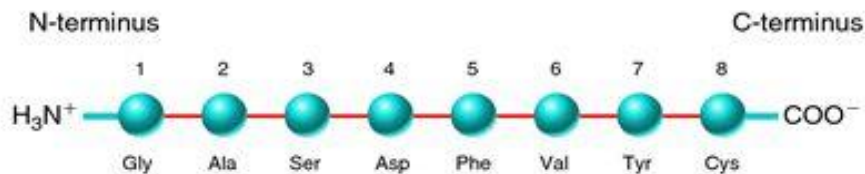


The amino end of a polypeptide is called the N-terminus.

The backbone of the polypeptide is highlighted in yellow.

The carboxyl end of a polypeptide is called the C-terminus.

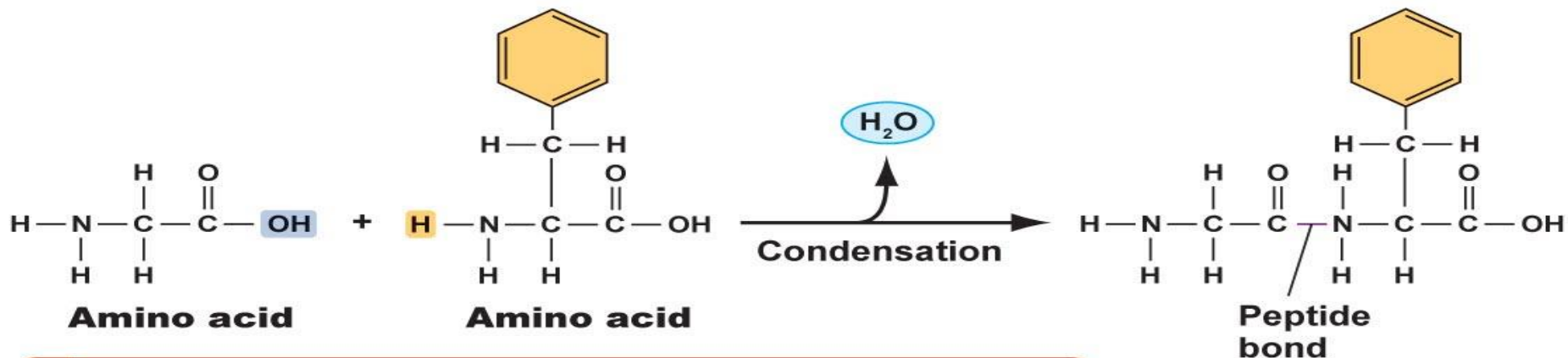
(b) Polypeptide—a linear chain of amino acids



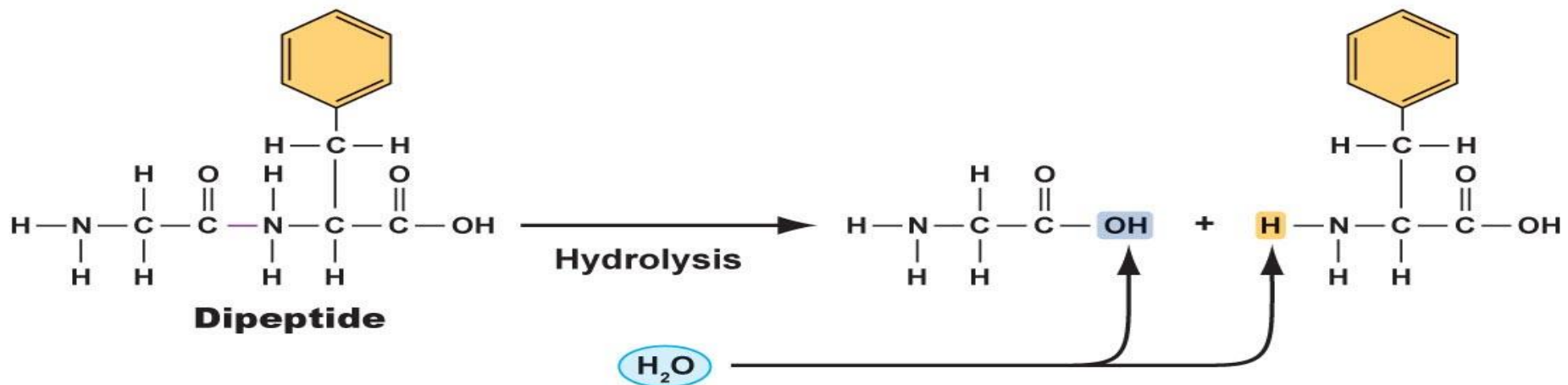
This is an octapeptide (8 amino acids).

(c) Numbering system of amino acids in a polypeptide

Condensation and Hydrolytic Reactions of amino acids/dipeptides



a A peptide bond forms by condensation when the acid group (COOH) and amine group of two different amino acids join and release a molecule of water.



b When peptide bonds are broken by hydrolysis, the hydroxyl group (OH) and hydrogen (H) from water are added.

Classification of Amino Acids

- Amino acids are most logically grouped according to the physicochemical properties of their side chains.
- They are also classified based on the nutritional needs

Classification Based on Nutritional Needs

Essential	Conditionally Non-Essential	Non-Essential
Histidine	Arginine	Alanine
Isoleucine	Asparagine	Asparatate
Leucine	Glutamine	Cysteine
Methionine	Glycine	Glutamate
Phenylalanine	Proline	
Threonine	Serine	
Tryptophan	Tyrosine	
Valine		
Lysine		

Classification based on the side chain

They are classified into:

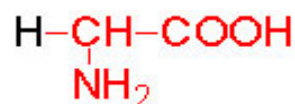
- Amino acids with non-polar side chains.
- Amino acids with aromatic side chains.
- Amino acids with uncharged polar side chains.
- Amino acids with positively charged R groups (basic amino acids).
- Amino acids with negatively charged R groups (acidic amino acids).

AAs with Non-polar Side Chains

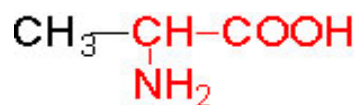
- The side chains cluster in the interior of the protein due to hydrophobicity.
- The side chain of **proline** and its α -amino group form a ring structure.
- **Proline** gives the fibrous structure of collagen, and interrupts the α -helices found in globular proteins.

AAs with Non-polar (Hydrophobic) R Groups

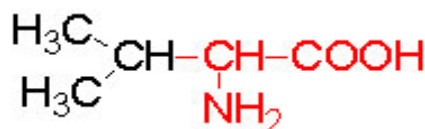
Glycine (Gly, G)



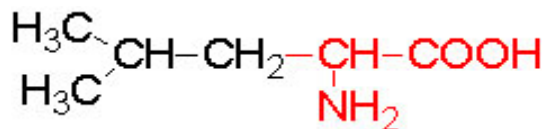
Alanine (Ala, A)



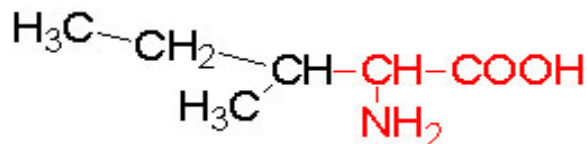
Valine (Val, V)



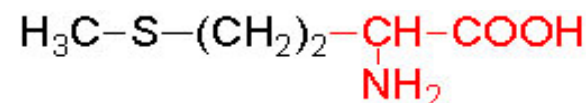
Leucine (Leu, L)



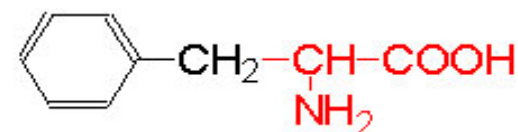
Isoleucine (Ile, I)



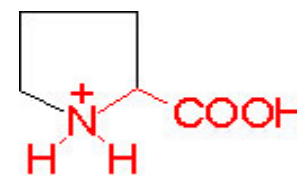
Methionine (Met, M)



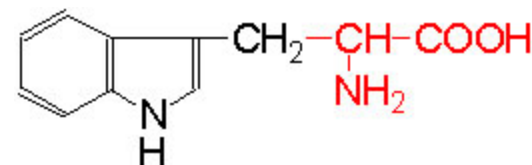
Phenylalanine (Phe, F)



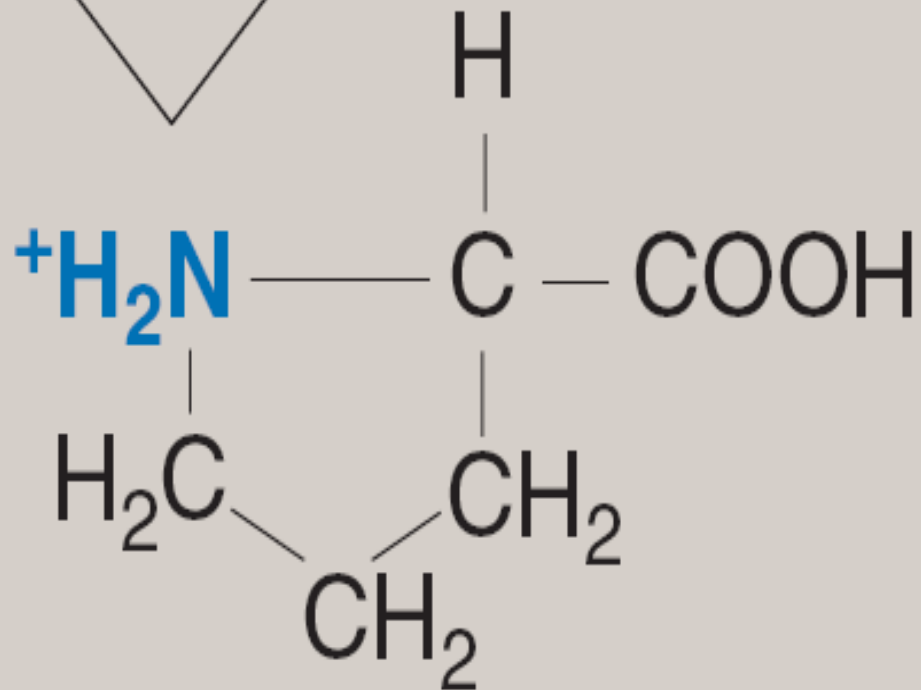
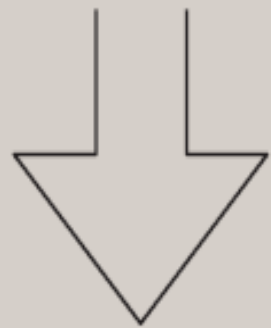
Proline (Pro, P)



Tryptophan (Trp, W)

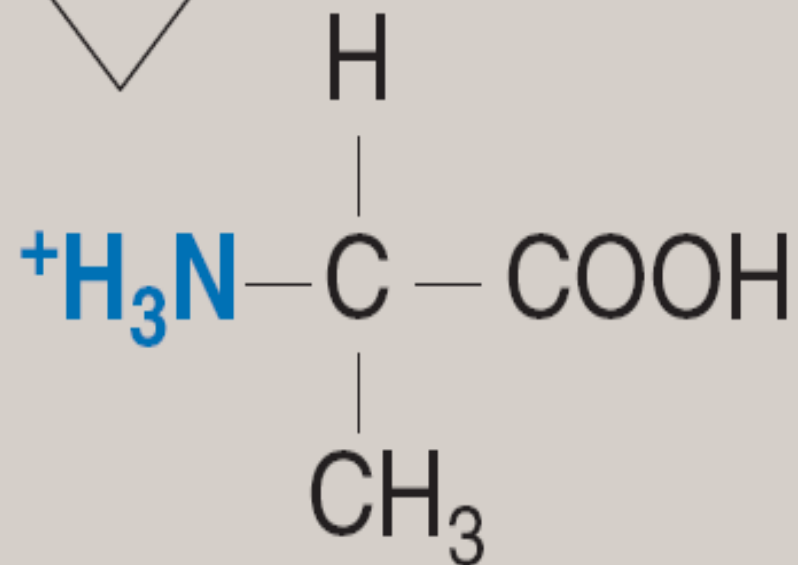
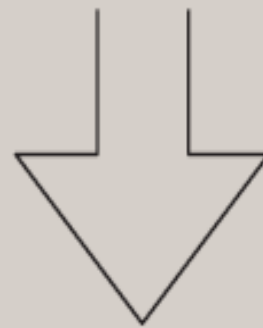


**Imino
group**



Proline

**Amino
group**

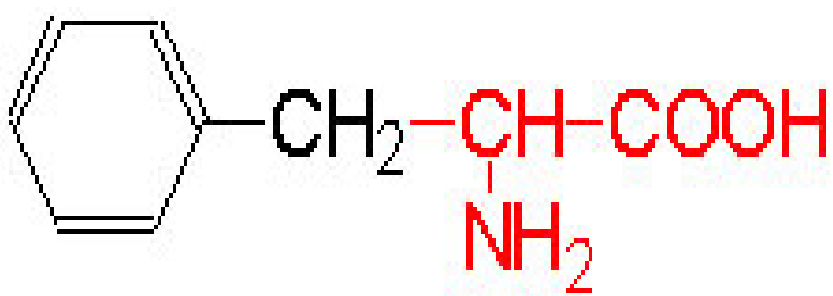


Alanine

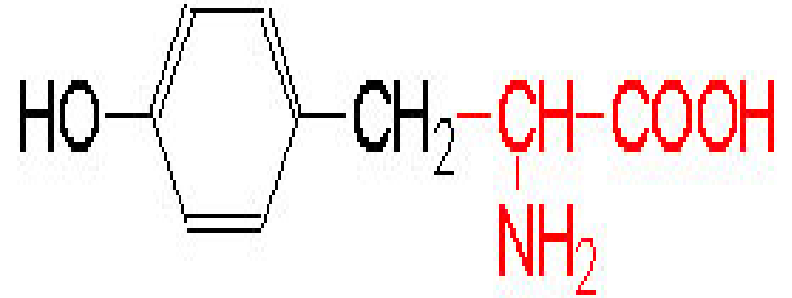
AAs with Aromatic (R) Groups

- Their aromatic side chains, are non-polar so that participate in hydrophobic interactions.
- Tyrosine is an important in some enzymes.
- Most proteins absorb light at a wavelength of 280 nm due to aromatic groups.
- A property exploited by researchers in the characterization of proteins.

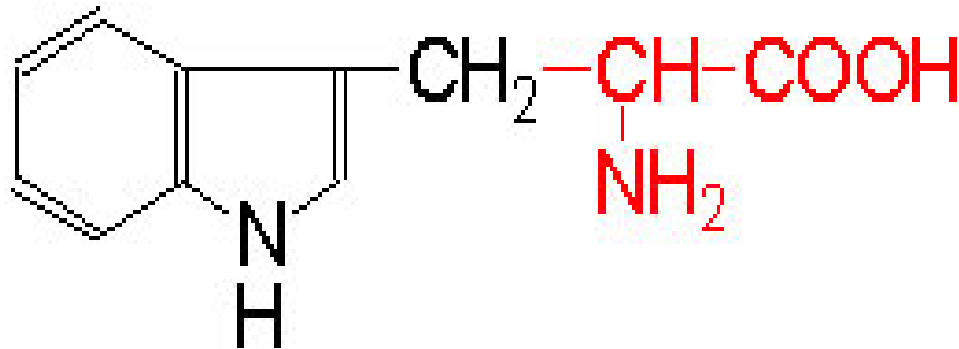
AAs with Aromatic (R) Groups



(Phenylalanine, Phe, P)



(Tyrosine, Tyr, Y)



(Tryptophan, Trp, W)

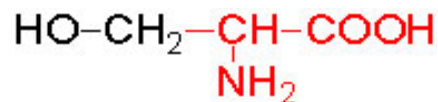
Structures of Amino Acids with Aromatic R-groups

AAs with Uncharged Polar Side Chains

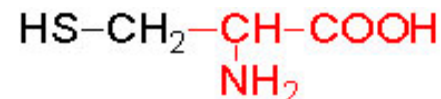
- More hydrophilic because they form hydrogen bonds with water.
- They include: **serine, threonine, cysteine, asparagine, and glutamine.**
- **Cysteine** contains a **sulfhydryl group** (-SH), an important component of the active site of many enzymes.
- Two cysteine molecules can be linked via disulphide bond (-S-S-) to form a dimer **cystine**.

AAs with Polar (Hydrophilic) R Groups

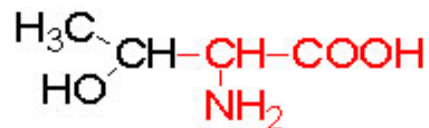
Serine (Ser, S)



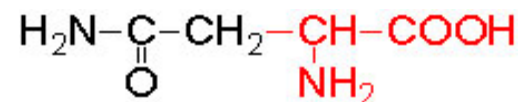
Cysteine (Cys, C)



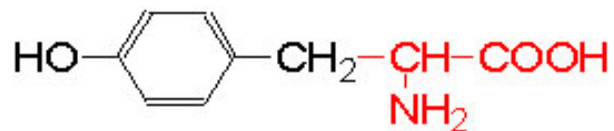
Threonine (Thr, T)



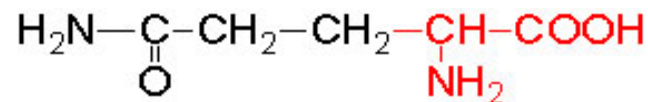
Asparagine (Asn, N)



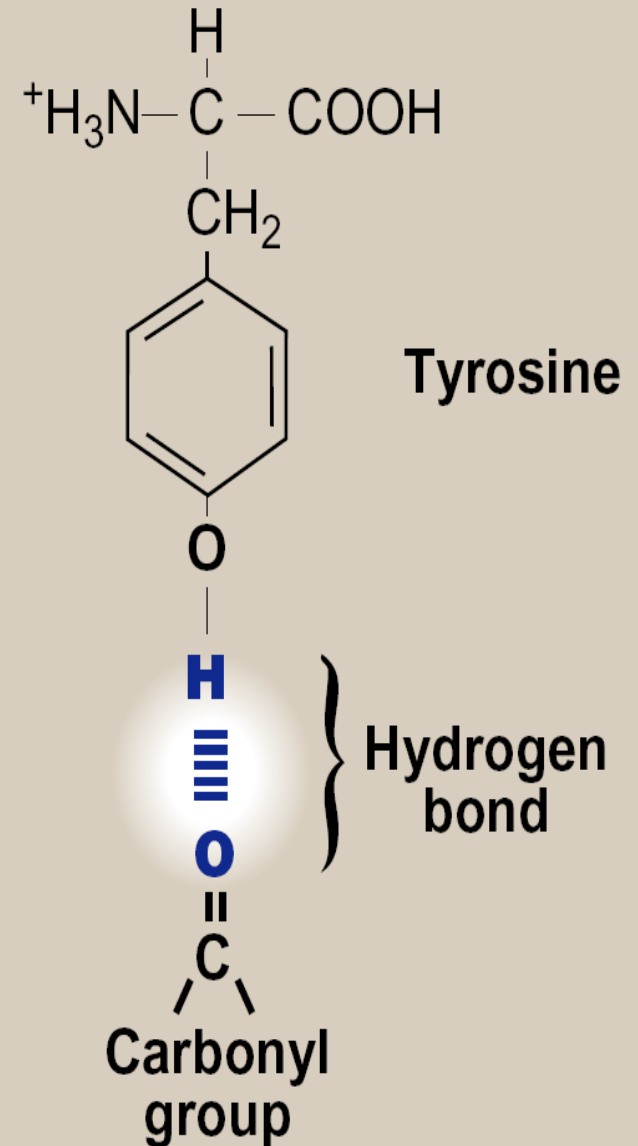
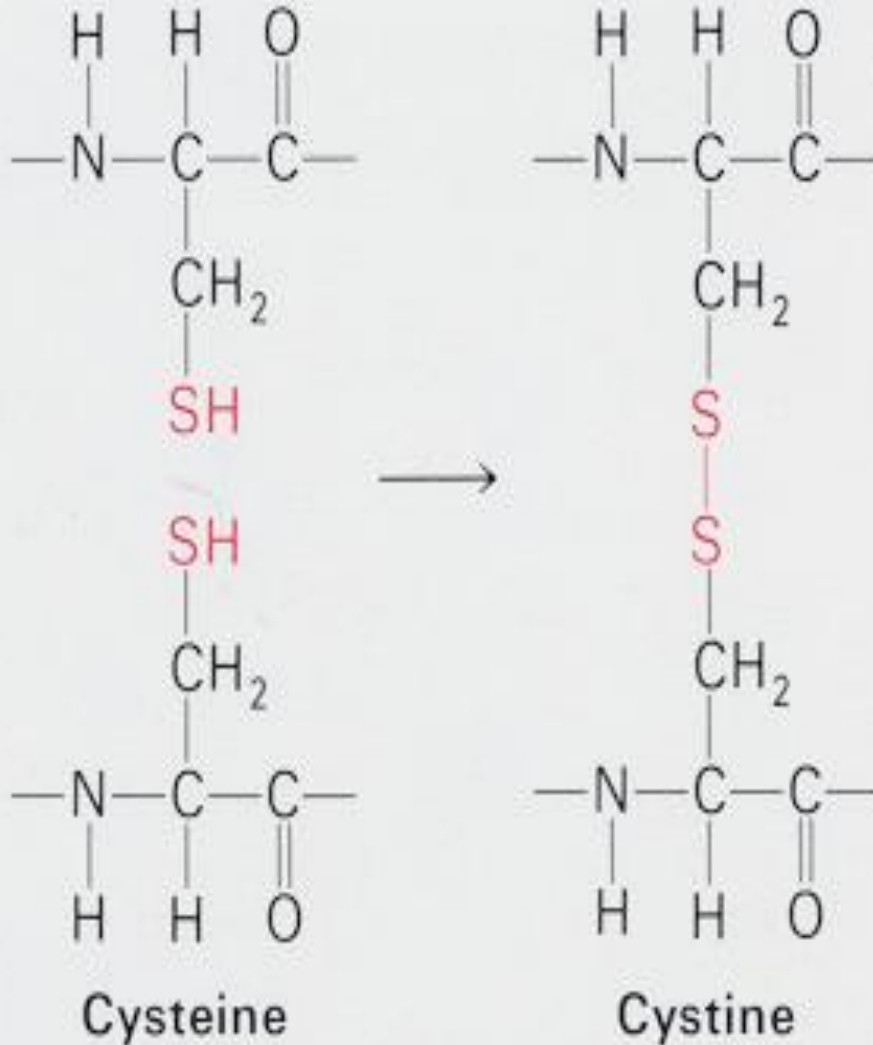
Tyrosine (Tyr, Y)



Glutamine (Gln, Q)

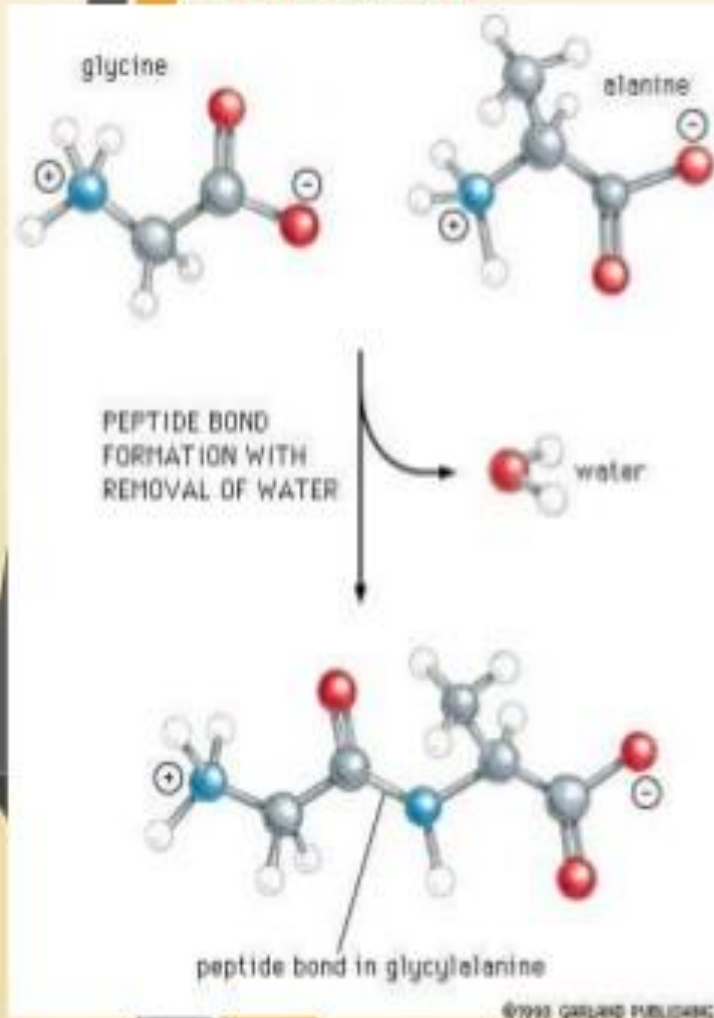


Role of side chains in bond formation

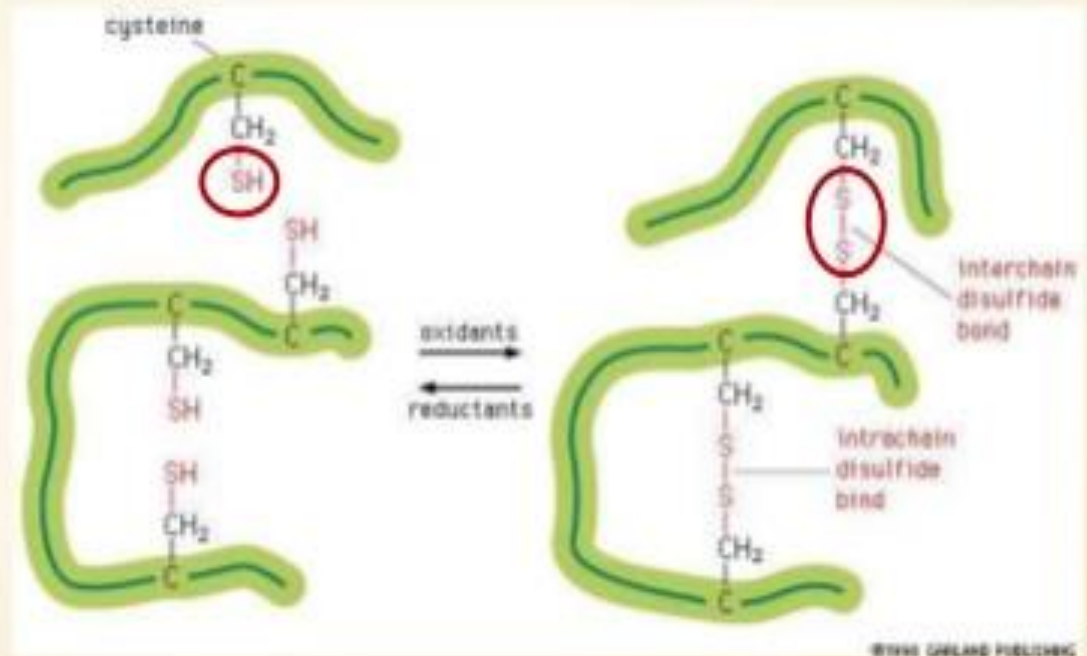


Amino acids side chains & bonds in Proteins

Peptide bond



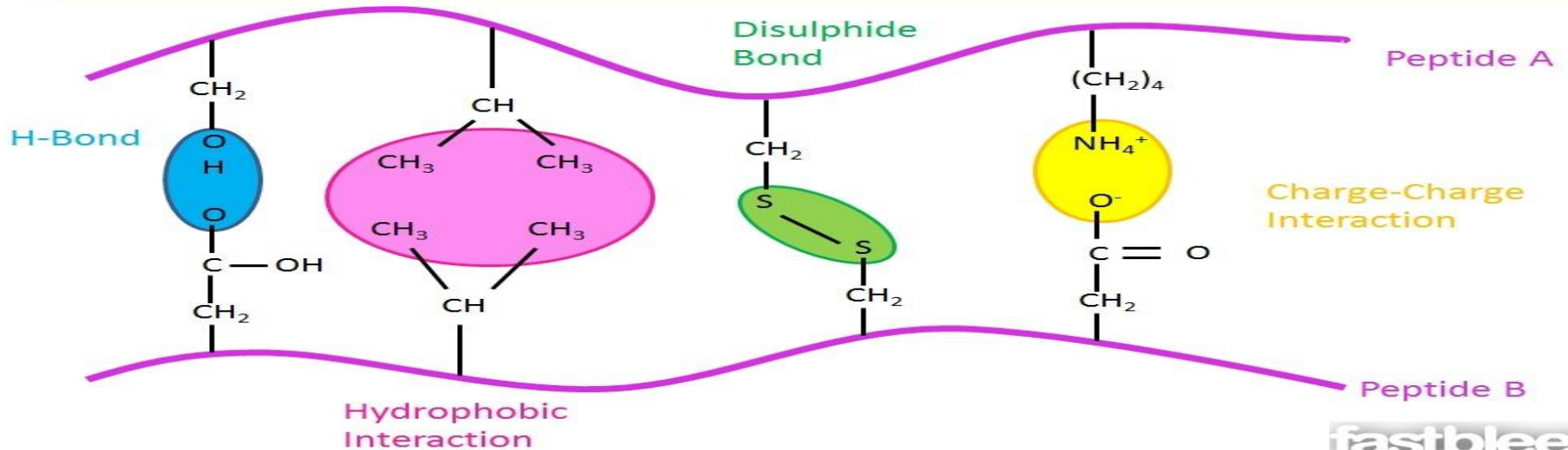
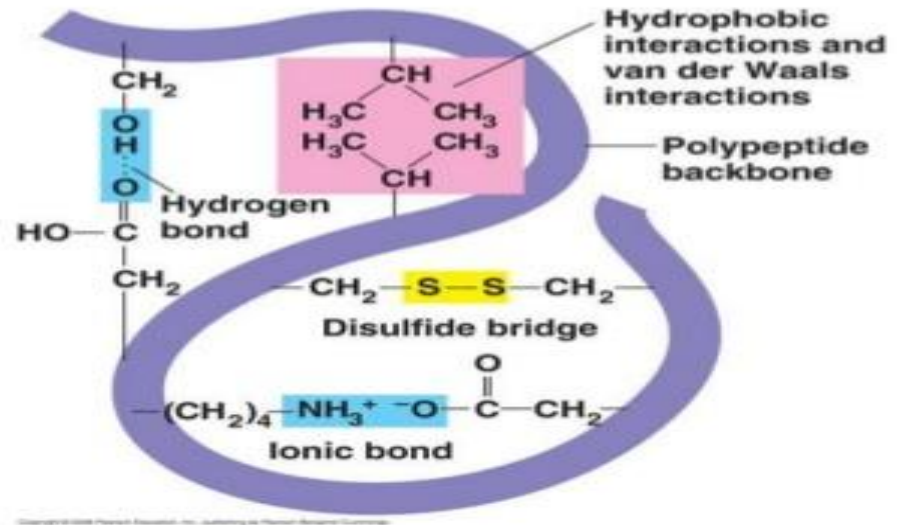
Covalent bond



Disulphide bond

INTERACTIONS STABILIZING 3^o STRUCTURE

- This final shape is determined by a variety of bonding interactions between the "side chains" on the amino acids.
- Hydrogen bonds
- Ionic Bonds
- Disulphide Bridges
- Hydrophobic Interactions:

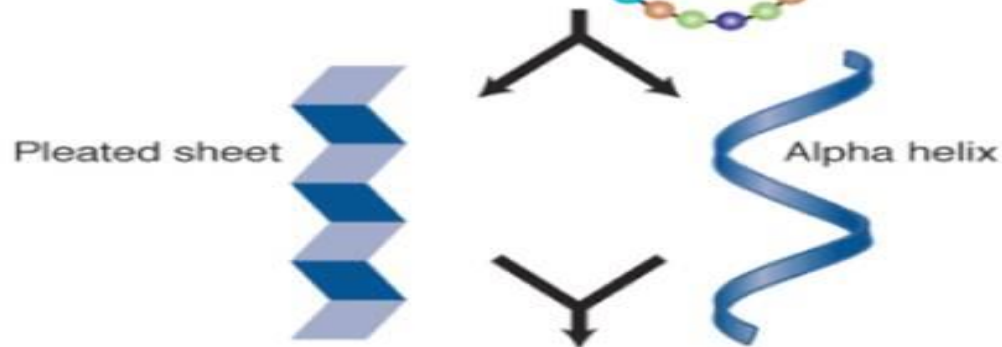


Levels of Protein Organizations

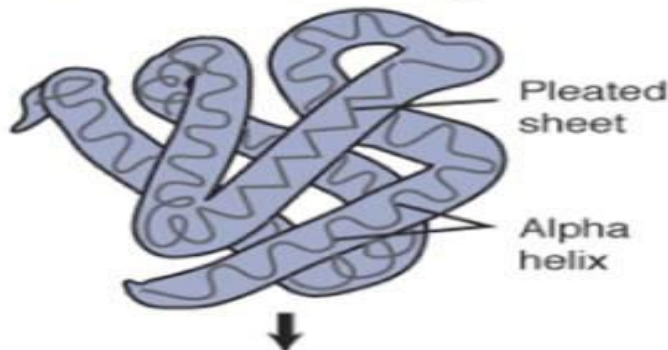


Levels of protein organization

Primary protein structure
is sequence of a chain of amino acids



Secondary protein structure
occurs when the sequence of amino acids
are linked by hydrogen bonds



Tertiary protein structure
occurs when certain attractions are present
between alpha helices and pleated sheets.



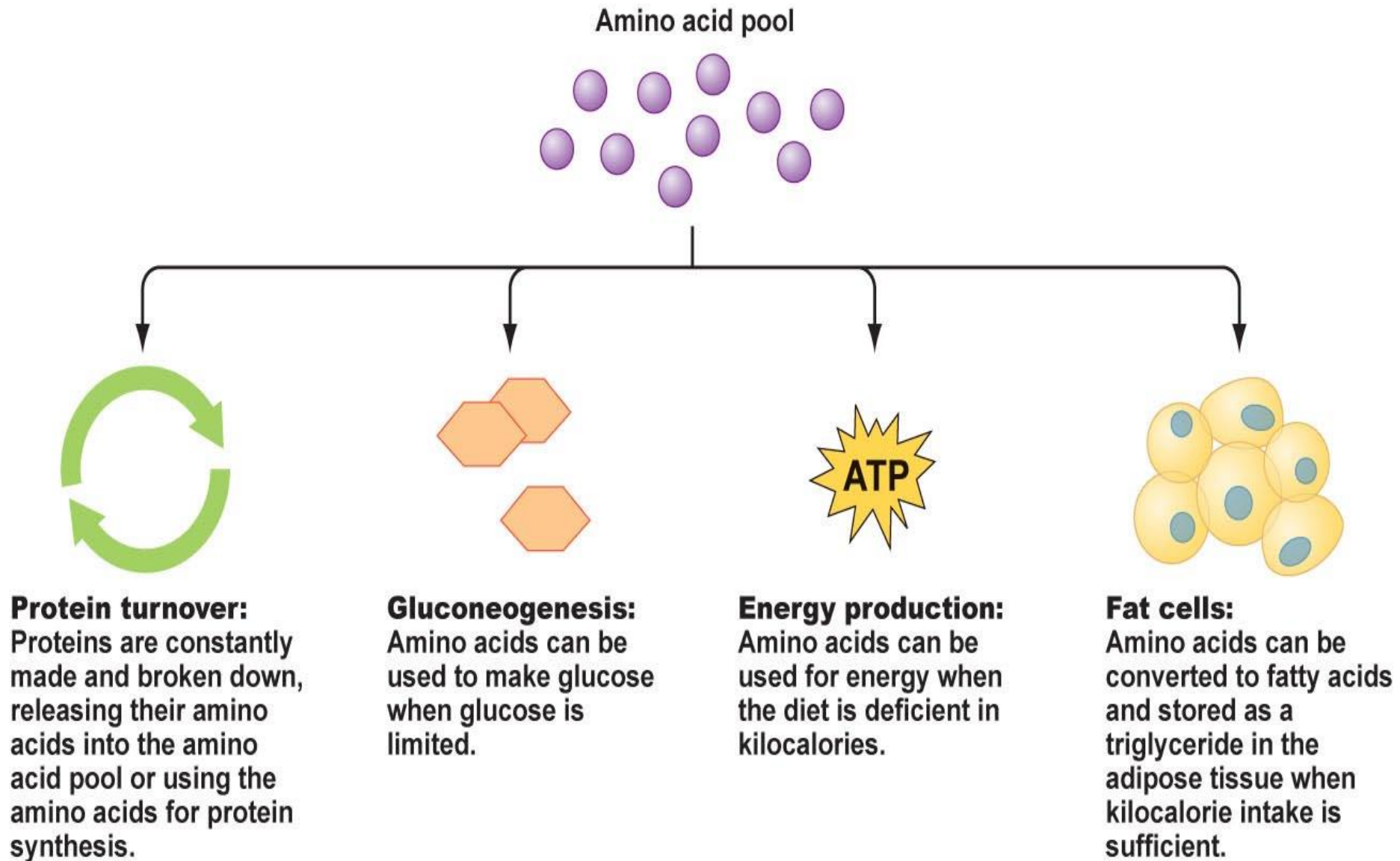
Quaternary protein structure
is a protein consisting of more than one
amino acid chain.

Thank you

LECTURE 2

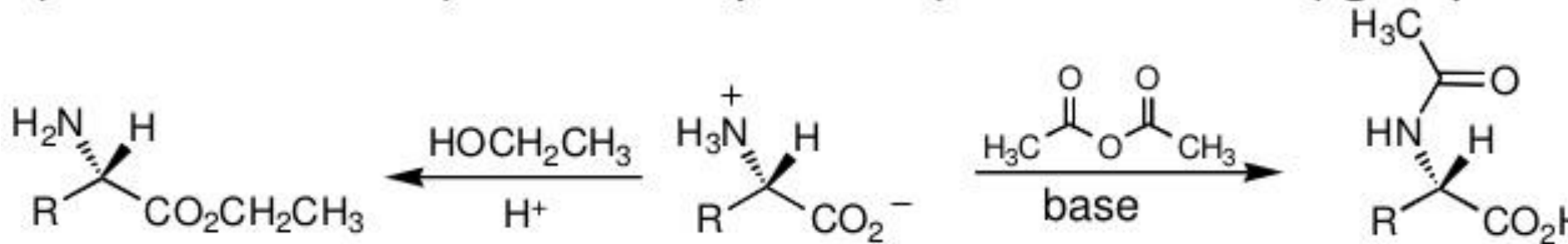
Nonpolar, aliphatic R groups	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{H} \end{array}$ Glycine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_3 \end{array}$ Alanine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH} \\ / \quad \backslash \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$ Valine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH} \\ / \quad \backslash \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$ Leucine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{S} \\ \\ \text{CH}_3 \end{array}$ Methionine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{CH}_3 \\ \\ \text{CH}_2 \\ \\ \text{CH}_3 \end{array}$ Isoleucine	Positively charged R groups $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH}_3^+ \end{array}$ Lysine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH} \\ \\ \text{C} = \text{NH}_2^+ \\ \\ \text{NH}_2 \end{array}$ Arginine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{C} - \text{NH} \\ \quad \backslash \\ \text{CH} \quad \text{CH} \\ \quad \backslash \\ \text{H} \quad \text{N} \end{array}$ Histidine
Negatively charged R groups	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2\text{OH} \end{array}$ Serine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{CH}_3 \end{array}$ Threonine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{SH} \end{array}$ Cysteine	Negatively charged R groups $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{COO}^- \end{array}$ Aspartate $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{COO}^- \end{array}$ Glutamate
Polar, uncharged R groups	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_2\text{N}^+ - \text{C} - \text{H} \\ / \quad \backslash \\ \text{H}_2\text{C} \quad \text{CH}_2 \end{array}$ Proline $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{C} \\ / \quad \backslash \\ \text{H}_2\text{N} \quad \text{O} \end{array}$ Asparagine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{C} \\ / \quad \backslash \\ \text{H}_2\text{N} \quad \text{O} \end{array}$ Glutamine	Nonpolar, aromatic R groups $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_5 \end{array}$ Phenylalanine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_4\text{OH} \end{array}$ Tyrosine $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{H} \\ \\ \text{CH}_2 \\ \\ \text{C} = \text{CH} \\ \quad \backslash \\ \text{C}_6\text{H}_4 \quad \text{NH} \end{array}$ Tryptophan

Metabolic Fate of Amino Acids



Reactions of amino acids

Amino acids will undergo reactions characteristic of the amino (amide formation) and carboxylic acid (ester formation) groups.



1- Reactions due to COOH group:

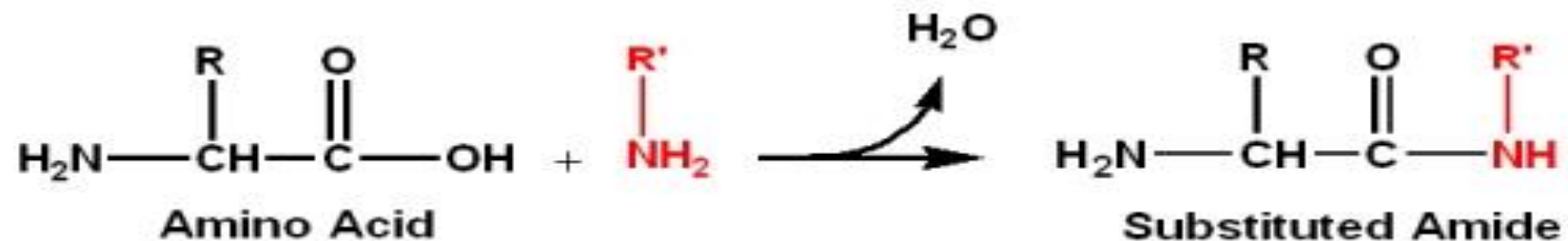
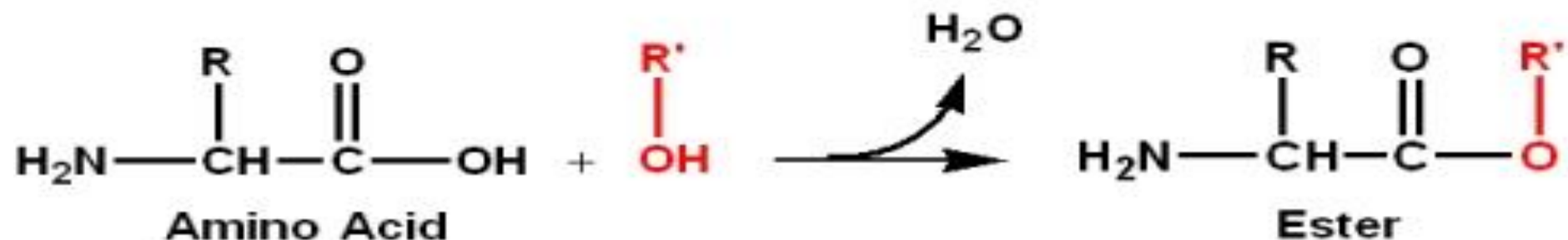
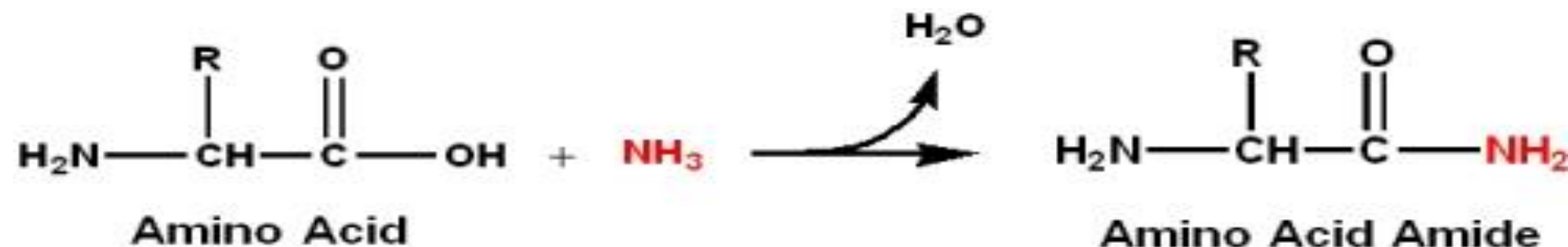
- Salt formation with alkalis, ester formation with alcohols, amide formation with amines and decarboxylation

2- Reactions due to NH_2 group: Deamination & reaction with ninhydrin

Ninhydrin reagent reacts with amino group of amino acid yielding colored product. The intensity of blue color indicates quantity of amino acids present.

- Ninhydrin can react with imino acids as proline and hydroxy proline but gives **yellow color**

Reactions due to carboxylic group



Decarboxylation of amino acids

Amino acids undergo α -decarboxylation to form their corresponding amines

Example:

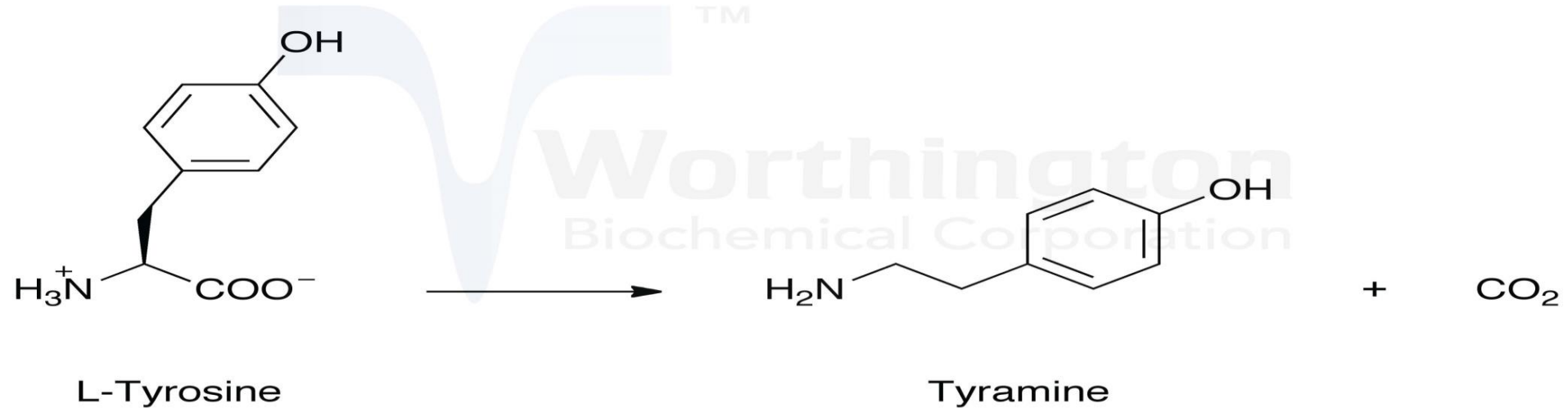
Histidine to histamine

Tyrosine to tyramine

Tryptophan to tryptamine

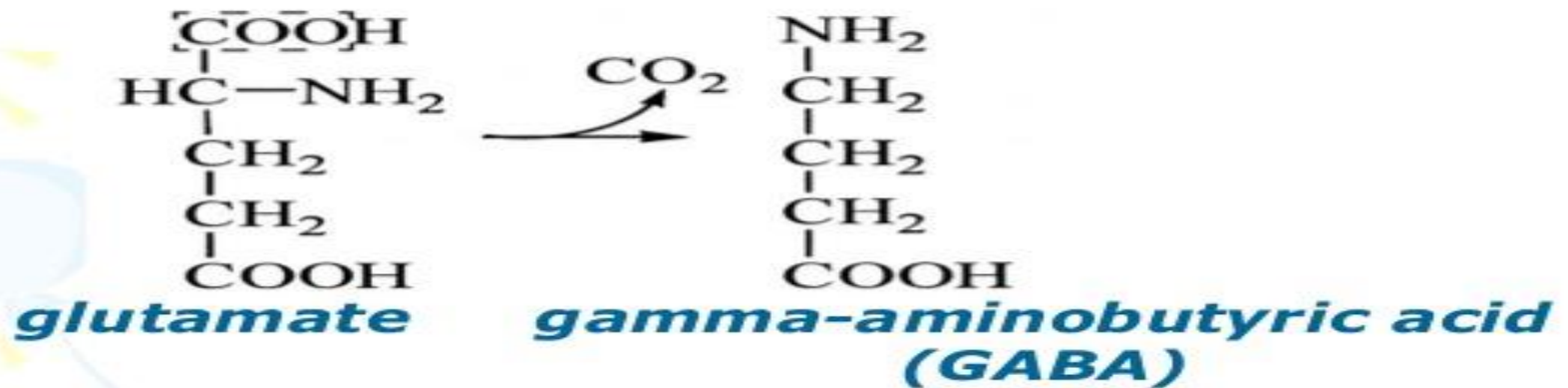
Glutamic acid to gamma aminobutyric acid (GABA)

Tyrosine Decarboxylase

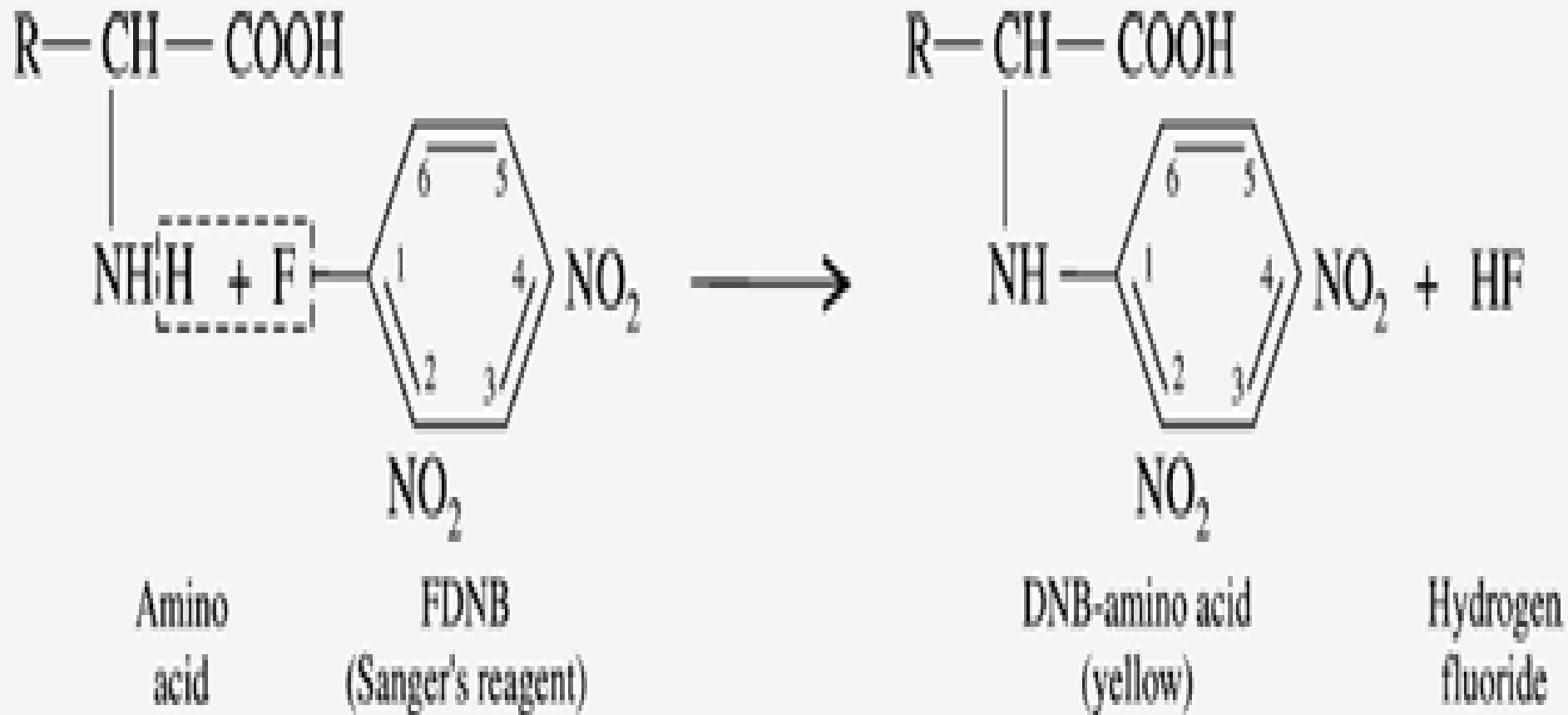


Significance of amino acid decarboxylation

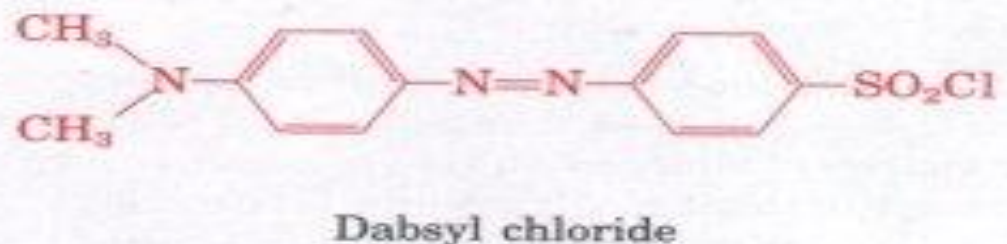
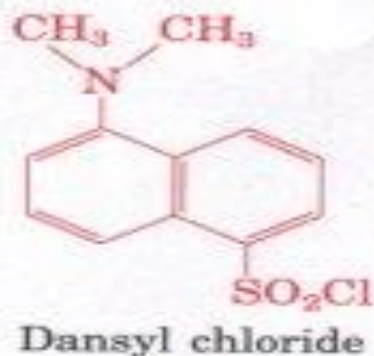
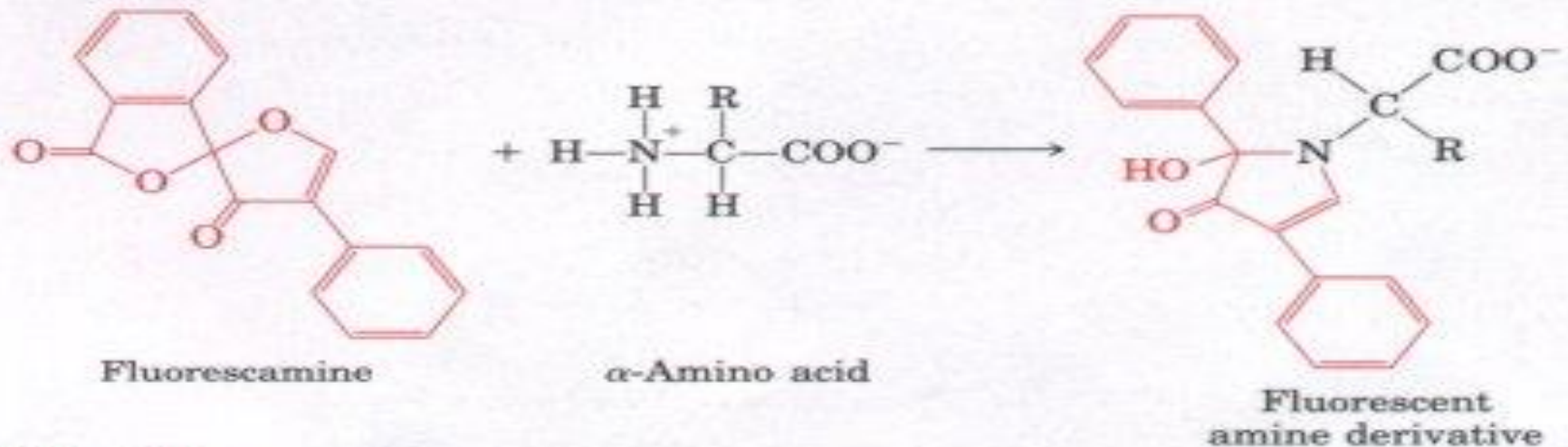
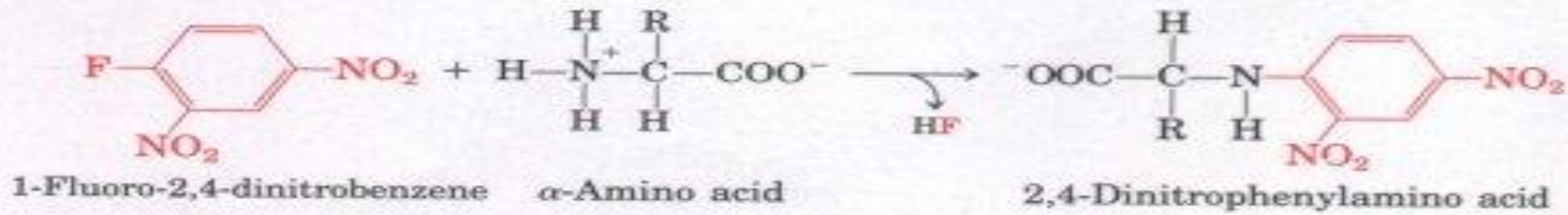
1. Formation of physiologically active compounds



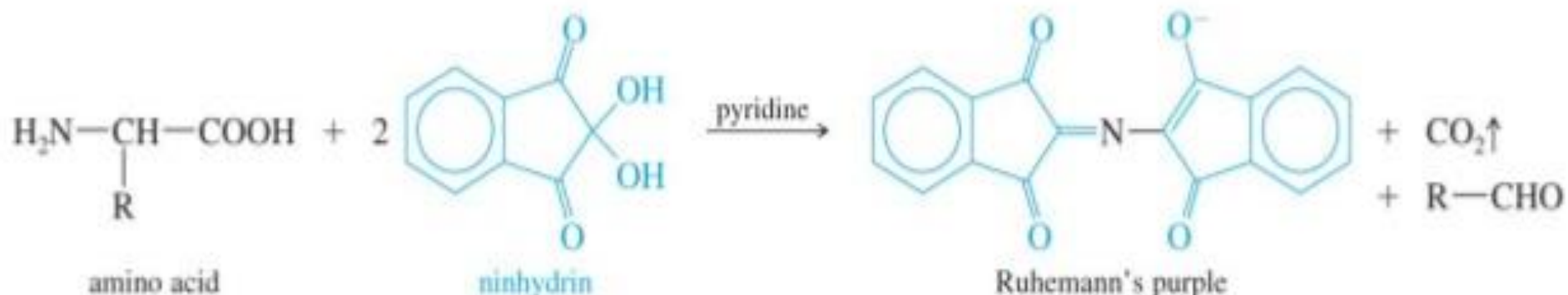
Reactions due to amino group



Other reagents that react with amino acids like Sanger's reagents



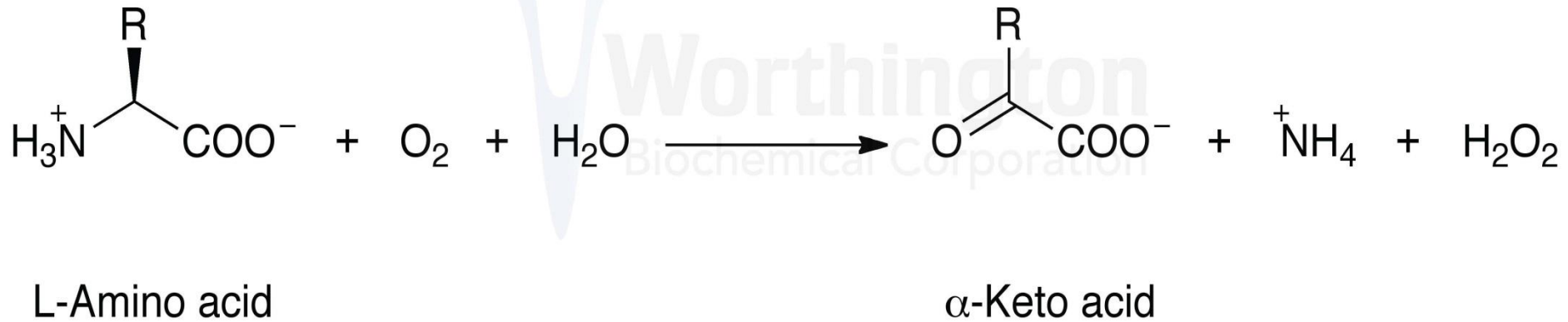
Reaction with Ninhydrin

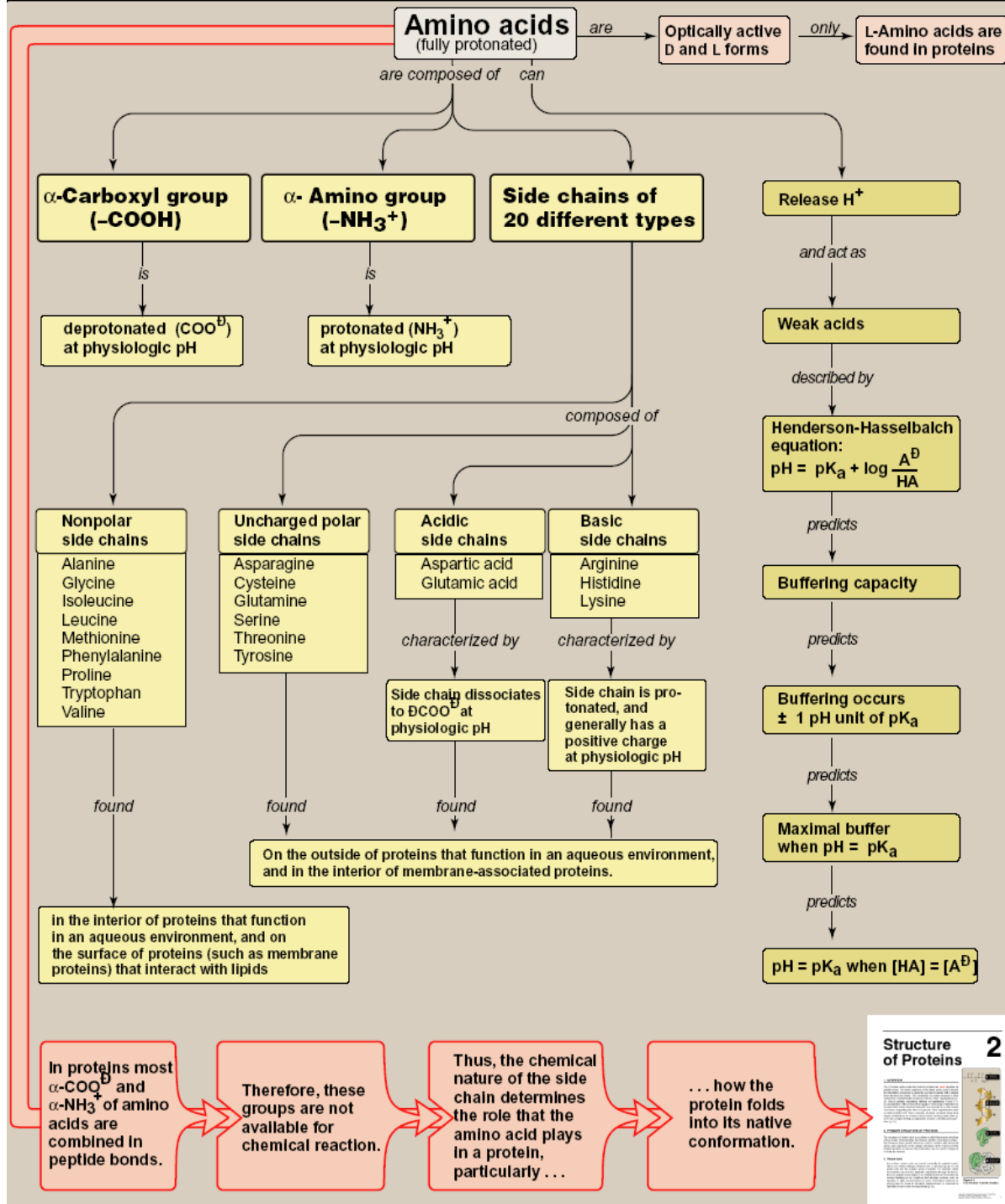


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- Used to visualize spots or bands of amino acids separated by chromatography or electrophoresis.
- Deep purple color formed with traces of any amino acid.

L-Amino Acid Oxidase





Summary

- Each AA has an **α -carboxyl** and an **α -amino** group (except Pro, has an **imino** group)
- At physiologic pH: α -carboxyl $\rightarrow$ COO^- , α -amino $\rightarrow$ NH_3^+
- Each aa contains one of 20 **side chains**
- The chemical nature of side chain determines function of AA in a protein, and classifies AA as **nonpolar, uncharged polar, acidic, or basic**.
- The **α -carbon** of each aa (except Gly) is attached to 4 different chemical groups i.e., a **chiral** or **optically active** carbon atom.
- Only L-form of AA's is found in proteins synthesized by human body.

The end

of

this section